

YOUR COLLECTION, ON THE WALL. EVERY CARD OPENS LIKE A DOOR.

SWINGRACK is a 3D-printed wall rack for carded die-cast cars. Each card gets its own swing-out ledge. The ledges are stacked and staggered so every name stays readable, and you can take any one card out without touching the others.



6-car kit on the 40 cm strip, closed: each row steps forward 7 mm, so every car and name stays in view.



Early printed prototype holding real carded cars (photo by the designer).

THE PROBLEM

- Carded cars end up in boxes or on peg hooks, where they hang behind each other and the backs get bent.
- Getting one card off a crowded hook means unhooking the cards in front of it.
- Most shelf displays are built for loose cars, not for cards with a blister on the front.

THE SOLUTION

- One ledge per card, hinged on a vertical pin. Lift it 1 mm, swing it out, slide the card out.
- A slot with a 1 mm front lip, a tall back wall and 14 mm corner posts holds the card upright, so it can't lean forward, while its name stays visible.
- Modular: one 40 cm wall strip carries up to 7 cars, 54 mm apart. It also prints as two halves that lock together.

HOW IT WORKS

01

Screw the strip to the wall

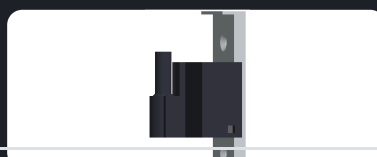
Four 5 mm countersunk screws.
40 cm strip, racks 54 mm apart.



02

Drop a clip on a hook lip

Drop it onto a lip, then push the lock pin through: now it can't lift off. To remove it, pull the pin, then lift and pull.



03

Drop the ledge on the pin

Slide a card into the slot. A small bump clicks the ledge shut and the solid clip block keeps it at 0°.



3

printed part types: strip,
clip, ledge

0

support material needed to
print

2-6

car kits on one 40 cm strip

~6 mm

lift and pull to take a clip
off

ENGINEERING THAT SURVIVED THE TEST PRINTS

Hooks that can't snap

The hook lips run across the full 30 mm width of the strip and print flat with a 45° under-fillet, so there is nothing thin to break off. The earlier pegs broke where they met the strip.

No sagging ledge

The ledge sits on a wide solid footing, with an 8 mm pin running its full height. The clip's solid block stops it at 0°, and a bump-and-dimple detent keeps it closed.

40 cm strip, any printer

Big bed: print it in one piece. Smaller bed: print two halves (180 + 234 mm) and press them together; a C-shaped interlock locks them.

Hook + lock pin, no tools

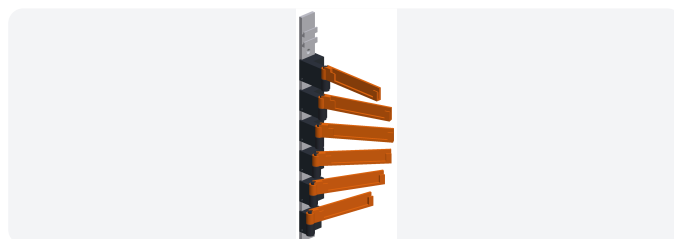
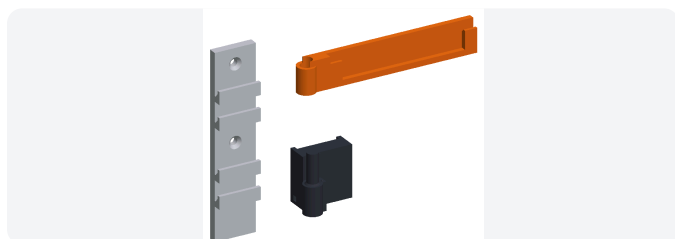
A 30 mm finger drops behind a lip, and a small lock pin pushed through the clip sits under the lip below, so the clip can't lift off when you open a ledge. Side cheeks stop it twisting.

Names always visible

Each rack sits 7 mm further from the wall than the one below it. The 1 mm front lip leaves the card title in view.

Prints on any FDM printer

Every part was checked for overhangs: there are only two short bridges and no supports. PLA or PETG.



KIT LINE-UP

Kit	Wall strips	Hinge clips	Ledges	Stack height	Est. filament*
2 cars	1 (40 cm)	2 (depth 0–1)	2	109 mm used	≤ 185 g
3 cars	1 (40 cm)	3 (depth 0–2)	3	163 mm used	≤ 230 g
4 cars	1 (40 cm)	4 (depth 0–3)	4	217 mm used	≤ 275 g
5 cars	1 (40 cm)	5 (depth 0–4)	5	271 mm used	≤ 325 g
6 cars	1 (40 cm)	6 (depth 0–5)	6	325 mm used	≤ 380 g

*Upper bound from STL volume printed 100% solid in PLA; real prints with normal infill use less. Fits cards about 105 × 165 mm with a blister up to 42 mm tall.

WHO IT'S FOR

- Collectors who keep their cars carded.
- Hobby and toy shops that want a neat counter or wall display.
- Makers who buy printable files on Printables, Etsy and similar marketplaces.

NEXT STEPS

- Test print the 40 cm strip, clip and ledge, then do load and drop tests with full cards.
- Launch the STL files and printed kits. Set prices after the cost test.
- Add-ons: colour sets, snap-on name plates, an 8-car kit.